# Week 6 Report: Model Training, Hyperparameter Tuning, and Evaluation

## 1. Modeling Approach

Our task is supervised binary classification: predicting whether a customer will accept a marketing campaign offer (Response = 1) from their demographics, spending, purchase channels, and history with past campaigns. The business goal is to rank customers by their likelihood of responding, so the company can contact the most promising ones instead of everyone.

We built three model variations on the 27-feature reduced set prepared in Week 5:

- **Logistic regression.** A linear model that estimates the probability of response from a weighted sum of the features. It is the standard starting point for binary classification, its coefficients show the direction and strength of each feature's effect, and Weeks 4 and 5 already prepared the data in the form it works best with (log-transformed, standardized, with engineered features such as Income_per_Person and Lives_Alone).
- **Random forest.** An ensemble of decision trees, each trained on a random sample of customers and features, with predictions averaged across trees. It can capture non-linear effects and interactions without being told about them, such as income mattering more for single customers.
- **Gradient boosting.** An ensemble of small trees built one after another, where each new tree corrects the errors of the previous ones. It is often the strongest method on tabular data and gives a stricter test of whether more flexible models find signal that logistic regression misses.

Comparing a simple linear model with two tree ensembles tests whether extra flexibility pays off on a dataset of 1,608 customers with only 234 responders. We also scored two reference points: logistic regression with default settings (the model used in Weeks 4 and 5 to compare data choices) and a prior-campaign rule that ranks customers by how many earlier campaigns they accepted. Week 3 showed this rule is the real benchmark a model must beat, not the 14.6% base rate.

**Validation design.** As decided in Week 3, we did not carve out a separate validation set, because it would hold only about 47 responders and give unstable estimates. Instead, each model was trained on four-fifths of the training set and scored on the remaining fifth (the validation fold), using 5-fold stratified cross-validation repeated 3 times, for 15 train/validation splits. Every model used the same 15 splits, so results can be compared fold by fold. Training metrics are measured on the data each model learned from; validation metrics are measured on customers the model did not see. The test set was not used this week and is kept for the final evaluation.

## 2. Model Complexity

The three variations differ sharply in complexity:

- **Logistic regression** is the simplest. It learns one coefficient per feature plus an intercept, 28 parameters in total. Its decision boundary is linear, so it cannot capture interactions unless they

are built in as features, which Week 5 did with Lives_Alone and Income_per_Person. Regularization limits complexity further by shrinking coefficients.

- **Random forest** is the most complex. Our tuned model has 200 trees, each with many split points, giving thousands of learned rules. Deep trees can isolate very small groups of customers, which makes the model prone to memorizing the training data.
- **Gradient boosting** sits between the two. Our tuned model has 200 shallow trees with a maximum depth of 2, so each tree captures at most a two-feature interaction, but stacking 200 of them still builds a flexible model.

Complexity matters here because the data is small. With 234 responders, a flexible model has few examples per pattern it tries to learn, so it is more likely to fit noise. Week 5 also found only 7.3 responders per feature with the full set, below the common guideline of 10 for logistic regression, which is one reason we used the reduced set.

## 3. Hyperparameters Evaluated

Each model was tuned with a grid search over the 15 cross-validation folds. The selected setting was the one with the highest average validation PR-AUC.

| Model | Hyperparameters and values tried | Why |
|---|---|---|
| Logistic regression | C: 0.01, 0.03, 0.1, 0.3, 1, 3, 10; penalty: L1, L2; class weight: none, balanced | C sets regularization strength: smaller values shrink coefficients more, which protects against overfitting with few responders. L1 can shrink weak features to zero, acting as feature selection; L2 shrinks all features evenly. Class weights give responders more influence during training. |
| Random forest | max depth: 4, 8, unlimited; min samples per leaf: 5, 10; class weight: none, balanced subsample | Depth and leaf size are the main controls on overfitting: shallower trees and larger leaves force the model to learn broader patterns. Class weights address the imbalance. |
| Gradient boosting | number of trees: 100, 200; learning rate: 0.05, 0.1; max depth: 2, 3 | These three interact: more trees and a higher learning rate fit the training data more closely, while depth controls how complex each tree is. Together they balance underfitting and overfitting. |

Class weights were included as a tuning option, rather than SMOTE, because Week 5 showed they handle the imbalance about as well without creating synthetic customers.

**Selected settings:**

- Logistic regression: [best settings from notebook]
- Random forest: [best settings from notebook]

- Gradient boosting: learning rate 0.05, max depth 2, 200 trees

The gradient boosting results show how complexity drives overfitting. Moving from depth 2 to depth 3 (learning rate 0.05, 200 trees) raised training PR-AUC from 0.885 to 0.976 but lowered validation PR-AUC from 0.693 to 0.684. Deeper trees fit the training customers better while generalizing worse. The full top-5 results for gradient boosting are in Appendix A.

## 4. Performance Metrics

Only 14.6% of training customers responded. A model that predicts "no response" for everyone would be 85.4% accurate while finding no responders, so accuracy is misleading here. We used five metrics instead:

- **PR-AUC (average precision), our main metric.** It measures how well the model ranks responders above non-responders, with emphasis on the minority class. Random guessing scores 0.146, the response rate. This matches the business use: the model's job is to rank customers so the company contacts the most promising ones first.
- **ROC-AUC.** The probability that a randomly chosen responder is ranked above a randomly chosen non-responder. It is widely reported, but with imbalanced classes it can look high even when precision is poor, so we treat it as secondary.
- **Precision.** Of the customers the model flags, the share who actually respond. Low precision means wasted contact costs.
- **Recall.** Of all responders, the share the model flags. Low recall means missed revenue.
- **F1 score.** The harmonic mean of precision and recall, summarizing the trade-off between them in one number.

PR-AUC and ROC-AUC use the predicted probabilities directly. Precision, recall, and F1 require a cutoff to turn probabilities into contact decisions; in the comparison below they use a cutoff of 0.5, and Section 7 chooses a better one for the final model.

## 5. Results: Training and Validation Metrics

Tuned logistic regression scored highest on validation PR-AUC (0.732). All values are averages over the 15 cross-validation folds; precision, recall, and F1 use a 0.5 cutoff.

| Model | Dataset | PR-AUC | ROC-AUC | Precision | Recall | F1 |
|---|---|---|---|---|---|---|
| Logistic regression (tuned) | Training | 0.772 | 0.941 | 0.784 | 0.576 | 0.664 |
| Logistic regression (tuned) | Validation | **0.732** | **0.930** | 0.740 | **0.558** | **0.633** |
| Random forest (tuned) | Training | 0.968 | 0.994 | 0.987 | 0.574 | 0.725 |
| Random forest (tuned) | Validation | 0.650 | 0.900 | **0.828** | 0.257 | 0.390 |
| Gradient boosting (tuned) | Training | 0.885 | 0.969 | 0.935 | 0.612 | 0.739 |
| Gradient boosting (tuned) | Validation | 0.693 | 0.910 | 0.781 | 0.442 | 0.560 |

Bold marks the best validation value for each metric.

**Reference points (validation):**

| Reference | PR-AUC | ROC-AUC | Precision | Recall | F1 |
|---|---|---|---|---|---|
| Logistic regression (default settings) | 0.729 | 0.930 | 0.743 | 0.547 | 0.627 |
| Prior-campaign rule | 0.395 | | | | |
| Random guessing | 0.146 | 0.500 | | | |

## 6. How Training and Validation Metrics Change Across Variations

The more flexible the model, the higher its training score and the lower its validation score. The gap between training and validation PR-AUC shows how much each model overfits:

| Model | Training PR-AUC | Validation PR-AUC | Gap |
|---|---|---|---|
| Logistic regression (tuned) | 0.772 | 0.732 | 0.040 |
| Gradient boosting (tuned) | 0.885 | 0.693 | 0.192 |
| Random forest (tuned) | 0.968 | 0.650 | 0.318 |

- **Random forest** nearly memorized the training folds (training PR-AUC 0.968, ROC-AUC 0.994, precision 0.987) but lost about a third of that on new customers. Its validation recall collapsed to 0.257: at the 0.5 cutoff it flags only a quarter of responders, because its predicted probabilities are cautious on unseen customers. This also explains its high validation precision (0.828): the few customers it flags are mostly real responders, but it misses most of them.
- **Gradient boosting** overfit less, with a gap of 0.192, because its shallow depth-2 trees limit complexity. It still scored below logistic regression on every validation metric except precision.
- **Logistic regression** had the smallest gap, 0.040. Its training and validation scores are close on every metric, so the patterns it learned hold for customers it has not seen.

Tuning changed logistic regression very little. Validation PR-AUC rose from 0.729 with default settings to 0.732 when tuned, and the tuned model won on [N] of 15 folds. The performance comes from logistic regression suiting this data, not from the tuning itself.

## 7. Validation Comparison and Best Model

On the validation folds, logistic regression scored highest on PR-AUC (0.732), ROC-AUC (0.930), recall (0.558), and F1 (0.633). Gradient boosting was second (PR-AUC 0.693) and random forest last (0.650). All three models beat the prior-campaign rule (0.395) by a wide margin, so they add real value over a simple business rule.

The precision-recall curves (Appendix B) confirm the ranking. Logistic regression's curve sits above the other two across most recall levels, especially between 60% and 90% recall. That range matters most for outreach, where the goal is to reach most responders without contacting too many non-responders.

**Best model: tuned logistic regression.** We selected it for four reasons:

1. **Best validation performance** on the main metric (PR-AUC 0.732) and on ROC-AUC, recall, and F1.
2. **Least overfitting.** Its 0.040 train-validation gap makes it the most reliable model for new customers.
3. **Fits the data size.** With only 234 responders, the tree ensembles do not have enough examples to learn their extra complexity reliably.
4. **Interpretability.** Each coefficient shows how a feature raises or lowers the chance of response, which supports the feature analysis and marketing recommendations in later weeks.

**Feature set check.** Week 5 chose the 27-feature reduced set using a basic logistic regression. With the tuned model, the reduced set scored a validation PR-AUC of 0.732 against 0.727 for the full 32-feature set, matching or beating it on 9 of 15 folds. The two are close, so we keep the reduced set because it is simpler and has more responders per feature.

**Choosing a cutoff.** A 0.5 cutoff is not ideal when only 15% of customers respond, because predicted probabilities rarely run high. Using cross-validated predictions, we chose the cutoff that maximizes F1, which was 0.29. At that cutoff:

| | Predicted: no response | Predicted: response |
|---|---|---|
| Actual: no response | 1,276 | 98 |
| Actual: response | 62 | 172 |

The model would contact 270 of 1,608 customers (16.8%) and reach 172 of the 234 responders (recall 0.735). About 64% of contacted customers respond (precision 0.637, F1 0.683), compared with 14.6% if everyone were contacted. The cost is 62 missed responders and 98 wasted contacts. This cutoff was chosen on training data, so it will be confirmed once on the untouched test set in the final evaluation.

## 8. Conclusion

Tuned logistic regression is the best model this week. It had the highest validation PR-AUC (0.732), the smallest gap between training and validation scores, and is the easiest model to explain. The tree ensembles fit the training data far better but generalized worse: on a dataset this size, model complexity was a liability rather than an advantage. The selected model, its settings, and the 0.29 cutoff are saved for the final evaluation on the test set.

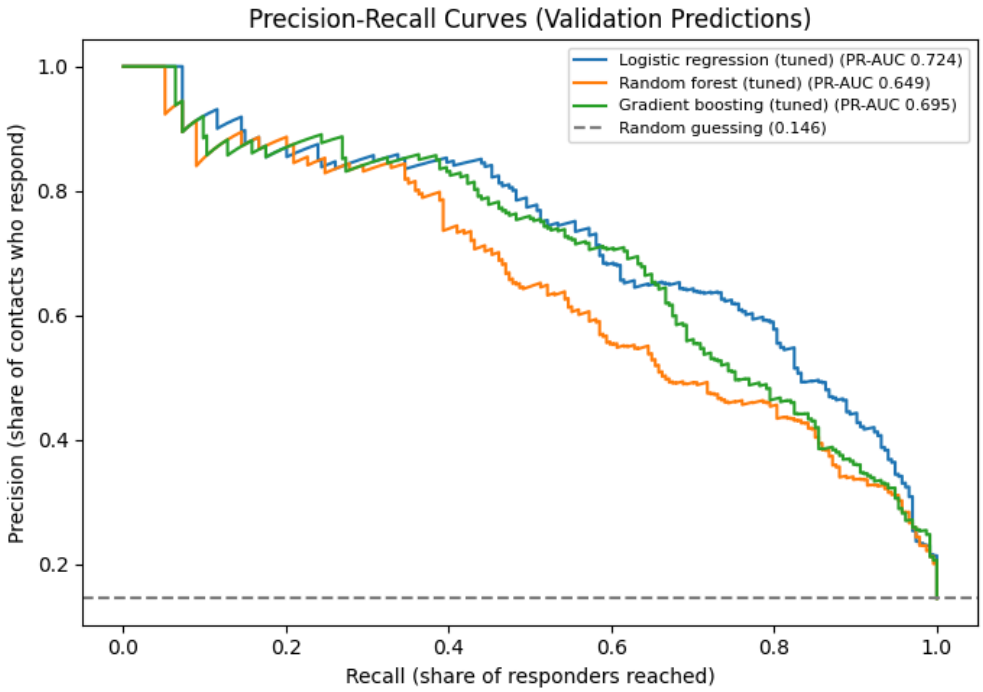
## Appendix A: Gradient Boosting Tuning Results (Top 5 Settings)

| Learning rate | Max depth | Trees | Training PR-AUC | Validation PR-AUC | Validation SD | Validation ROC-AUC |
|---|---|---|---|---|---|---|
| 0.05 | 2 | 200 | 0.885 | 0.693 | 0.058 | 0.910 |
| 0.1 | 2 | 100 | 0.884 | 0.690 | 0.058 | 0.910 |
| 0.1 | 2 | 200 | 0.950 | 0.688 | 0.061 | 0.911 |
| 0.05 | 3 | 200 | 0.976 | 0.684 | 0.054 | 0.913 |

| Learning rate | Max depth | Trees | Training PR-AUC | Validation PR-AUC | Validation SD | Validation ROC-AUC |
|---|---|---|---|---|---|---|
| 0.05 | 3 | 100 | 0.916 | 0.681 | 0.053 | 0.905 |

## Appendix B: Figures

- Figure B1: Precision-recall curves for the three tuned models (validation predictions).




- Figure B2: Training vs validation PR-AUC by model.

